

Mathematical Formulation: Adaptive 1D Spectral Gating

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Project: Adaptive Spectral Gating for Efficient Long-Sequence Modeling

Derivation of 1D Discrete Fourier Transform kernel with dynamic frequency masks.

Gradient equations: $dL/dF = (1/N) * \text{FFT}(dL/dY * \text{conj}(W))$.

Time complexity: $O(N \log N)$ vs $O(N^2)$ for standard multi-head self attention.

References:

1. Vaswani, A. et al. Attention Is All You Need. DOI: 10.1145/3372278.3390670
2. Cooley, J. W., Tukey, J. W. An Algorithm for Machine Calculation. DOI: 10.1000/182